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SHORT COMMUNICATION

City-Scale Seismic Risk Assessment Using Physics Based Earthquake Simulator: A Case Study of BHU Campus and Crossing Republik, Ghaziabad

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Abstract
Integrating key factors such as soil amplification, non-linear seismic response, and mass evacuation simulation is an effective way to comprehend a more holistic view of earthquake-induced damage at the urban level. Integrated Earthquake Simulation (IES) provides hazard and disaster predictions for various earthquake scenarios, using Geographical Information System data to generate digital city models. IES performs soil amplification, nonlinear seismic response analysis, and agent-based mass evacuation simulations to model earthquake events comprehensively. This study evaluates the seismic risk of the BHU Campus and Crossing Republik, Ghaziabad, under three earthquakes: the 1995 Kobe, 2001 Bhuj, and 2015 Nepal events. Digital twins of these areas were developed using shapefiles, incorporating attributes such as building height, construction material, structural system, and occupancy type. In the BHU campus, dominated by low-rise masonry structures, five-story or taller buildings exhibited inter-story drift ratios of 5 and 3.5% under the Kobe and Bhuj earthquakes, respectively, exceeding the allowable limit of 0.01h (h = building height), indicating significant seismic vulnerability. In the case of the 2015 Nepal Earthquake, the seismic responses increased by approximately 250% during the second wave, followed by a decrease in the third consecutive wave. The seismic response in Crossing Republik, Ghaziabad, shows that rooftop displacement exceeds the permissible limit of $h/500$, where h is the height of the structures), indicating a significant seismic risk in these areas.

Keywords Integrated earthquake simulation · Digital twin · Seismic response analysis · High-performance computing

Introduction

The seismic vulnerability of the Indian continent is very high as the Indian plate is driving into the Asian plate at an approximated rate of 47 mm per year Singh and Pandey [1]. As per the Geographical statistics, almost 54% of the land in India is vulnerable to earthquakes. United Nations report and a world bank estimate that by 2050, around 200 million city occupants in India will be exposed to earthquakes and storms?. Owing to high seismic activities in India, the new IS 1893:2022 has revised the micro-zoning and, introduced Seismic Zone VI, and eliminated Seismic Zone I.

Henceforth, all the buildings and civil infrastructures must be designed for earthquake resistance. Moreover, the seismic resilience of existing infrastructure must be done with the aim of strengthening these infrastructures. Individual structures are often considered in the common practice of earthquake resistance design as well as in seismic

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performance assessment of buildings. Contrary to this, an entire city or cluster of structures is damaged directly or indirectly in the event of a catastrophic earthquake. The damage report of the 2001 Bhuj Earthquake is the testimony of structures surviving the earthquakes but failing due to the pounding and falling of adjacent structures.

In view of this, it is essential to perform the seismic resilience assessment of clusters of structures or of entire cities, including the amplification of ground motion while traveling through the soil. This study is often known as urban simulation or Integrated Earthquake Simulation. Some of the significant aspects within the scope of the study of Integrated Earthquake Simulation are soil amplification analysis, structure response analysis, and mass evacuation simulations. These simulations also include the interaction between soil and structures. Miyamura et al. [2] performed a Large-scale seismic response analysis of a super-high-rise steel building incorporating Soil-structure Interaction using a supercomputer named K-computer. Ichimura et. al. [3] performed hybrid finite element modeling for seismic structural response analysis of the reinforced concrete structure using a non-linear elastoplastic concrete model. Chen et. al. [4] presented how to apply High-Performance Computing (HPC) for a Soil Dynamic Analysis (SDA) based liquefaction assessment for urban disaster mitigation and prevention. Vukobratovic et al. [5] studied the amplification effects due to the seismic response in alluvial soil. The authors performed equivalent linear one-dimensional ground response analysis on the dynamic model of soil in the frequency domain to investigate amplification effects at a site in the center of Novi Sad.

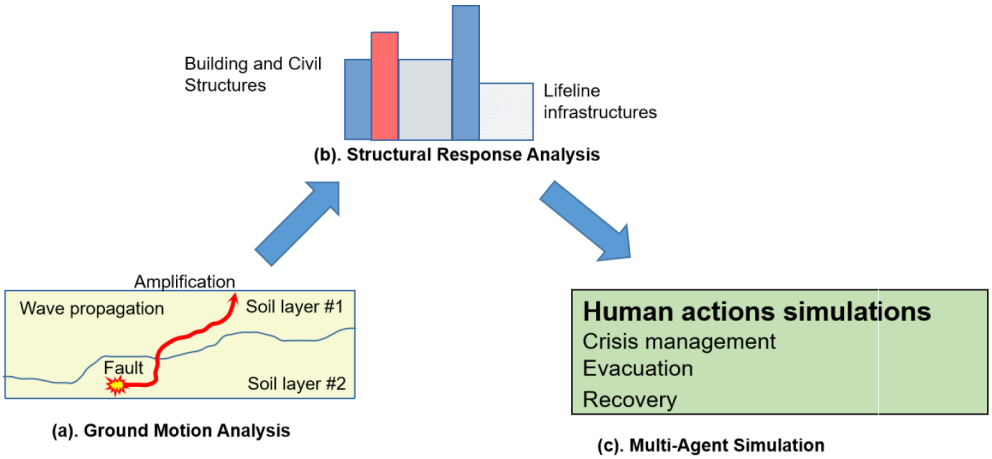
The above-mentioned literature on structure response analysis and soil amplification analysis incorporated high-fidelity numerical models but did not consider the impact of surrounding structures. To address this, Hori et. al. [6, 7]; Wijerathne, Hori, Ichimura, and Tanaka [8] developed salable parallel computing extensions for seismic response analysis and evacuation simulation modules for an Integrated Earthquake Simulator (IES) to simulate earthquake disasters in large urban areas. To further enhance the IES, Fujita et. al. [9] developed a rapid earthquake tragedy estimation system with fast urban earthquake simulation and interactive visualization. Furthermore, Sahin et. al. [10] extended the application of IES to study the seismic response of Istanbul city and developed a various module of IES on MATLAB. Taking a step further, Burton et. al. [11] have developed a tool integrating Performance-Based Engineering and Urban Simulation to model post-earthquake Housing Recovery. The efficacy of various intervention measures intended to facilitate post-earthquake housing recovery can be evaluated using simulation models to quantify their benefits and trade-offs. In the Indian context, seismic risk assessment of Indian cities have three levels of procedures, namely (1). Rapid Visual Screening (RVS) Jain, Mitra, Kumar, and Shah [12]; (US) [13] using visual inspection, (2). Simplified Vulnerability Assessment (SVA) using simplified engineering analysis based upon information collected through visual inspection R. Bhasker [14]; Sinha, Goyal, Shinde, Meena, and Sapre [15] and (3). Detailed Vulnerability Assessment (DVA) through detailed numerical analysis incorporating complex material behavior Baruah et al. [16]. While DVA is recommended for all prominent and lifeline structures, RVS and SVA are commonly used procedures. Moreover, seismic response analysis of Indian cities using high-fidelity models is not adopted because of computational cost involved. Although developing an integrated earthquake simulation tool, combining ground motion simulations and seismic response analysis, is definitely challenging but possible with the given state-of-the-art computational infrastructure in the country. It is, therefore, essential to perform the seismic performance analysis of Indian cities using existing tools and upgrade them if necessary.

In this pilot study, authors have adopted IES Hori et al. [6] for Indian cities and investigated seismic response analysis of the BHU campus and Crossing Republik Township, an integrated city project, for three earthquakes viz 1995 Kobe Earthquake, 2001 Bhuj Earthquake, and 2015 Nepal Earthquake. Digital 3-D models of these sites have been developed using shape-files and selected attribute data. Results show that in a hypothetical situation, had the 1995 Kobe earthquake occurred in Varanasi or Ghaziabad, both cities would have suffered significant damage. Had these sites been hit with an earthquake of the magnitude of the 2015 Nepal earthquake, most of the buildings, which are G+4 or higher, would have collapsed partially or totally.

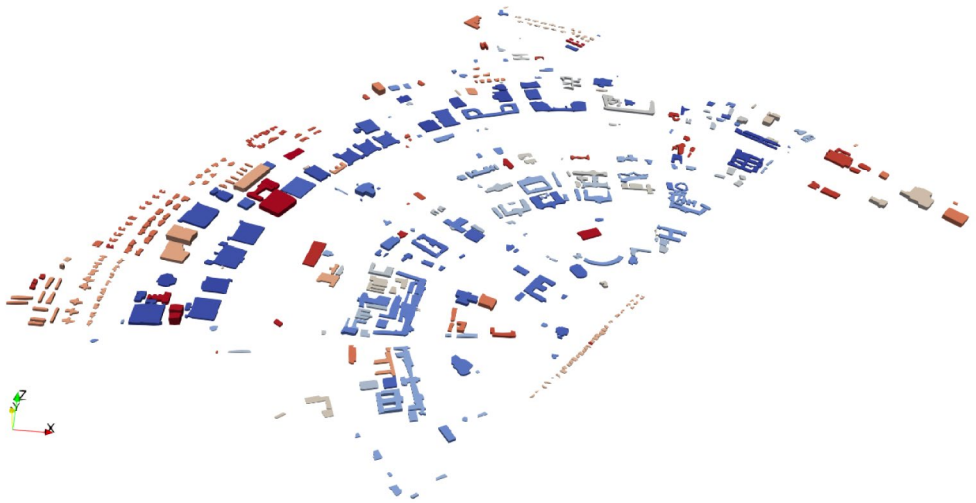
Integrated Earthquake Simulator (IES): An Overview

IES stands for Integrated Earthquake Simulation. IES realizes the estimation of earthquake hazards and disasters by analyzing all the involved processes. Various pairs of advanced numerical analysis methods and analysis models are used in IES. Each pair performs numerical analysis simulating a specific earthquake hazard and disaster process in a city and an urban area (see Fig. 1a). IES is a simulation system for the following three phases of an earthquake:

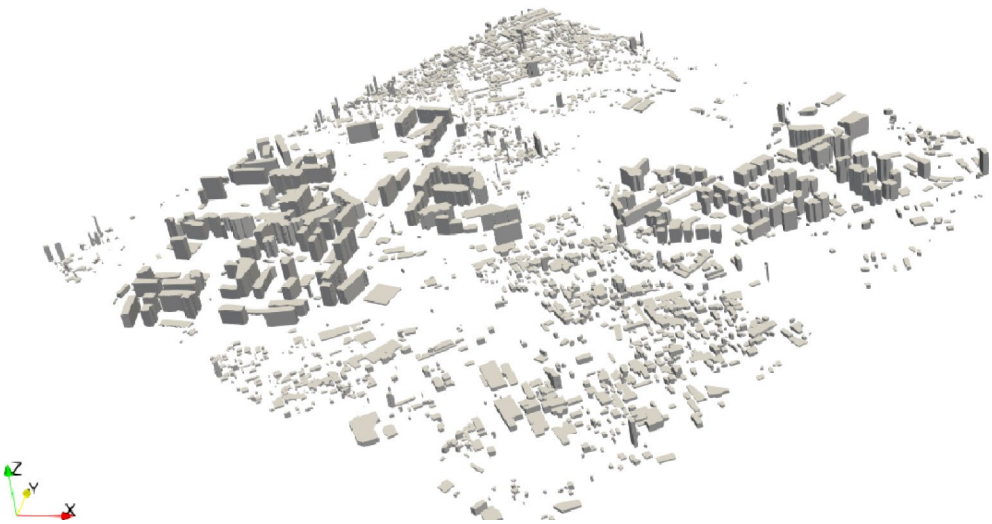
1. Generation and propagation of an earthquake wave.
2. Response of structures subjected to strong ground motion and damage to them.
3. Human or social actions against earthquake disasters.



(a) A schematic diagram of IES



(b) Constructed 3-D digital model of BHU campus



(c) Constructed 3-D digital model of Crossing Republik, Ghaziabad

Fig. 1 A schematic diagram of IES, digital twin of BHU campus and crossing Republik township

Mediator puts input data into a program of structural response analysis (SRA) to get the seismic response and simulation results for visualization. The SRA module can easily simulate (1). wooden structures, (2). reinforced concrete structures and (3), steel-reinforced structure.

Seismic Response Analysis

In IES, structural seismic response analysis is deep rooted with the concept of Newtonian mechanics based continuum mechanics. Lagrange for a linearly elastic structure, denoted by Ω, with displacement function, u is defined as follows:

$$L[\mathrm{u}] = \int_{\Omega} \frac{1}{2}\rho \dot{\mathrm{u}}.\dot{\mathrm{u}} - \frac{1}{2}\nabla \mathrm{u} : c : \nabla \mathrm{u} dv \tag{1}$$

where, ρ and c are density and elastic tensor, $\dot{(.)}$ and $\nabla(.)$ are temporal and spatial derivatives. Solution of this function is obtained using variational principal, $\delta \int L dt = 0$.

In IES, approximation of displacement function is based upon mass-spring system with emphasis upon modes of structural response. Thus, approximated displacement function is summation of temporal function (mode amplitude) and spatial function (mode shapes), which can be expressed as follows

$$\mathrm{u}(\mathrm{x}, t) = g(t)\mathrm{e} + \sum_{\alpha} U^{\alpha}(t)\phi^{\alpha}(\mathrm{x}) \tag{2}$$

where, g is input ground motion, e unit vector, U^{α} and ϕ^{α} are amplitude and shape of αth mode, respectively. Upon substituting Eq. 2 into Lagrange (Eq. 1) and applying variation of principal, following differential equation is derived:

Table 1 Attribute table sample

Building name	Floor area	Building type	Material	Floors	Year of construction
Faculty tower	1688	5	Masonry	10	2015
IIT new girls' hostel	4611.44	5	Masonry	7	2012
IIT guest house	1709.41	5	Masonry	2	2011
Limbdi hostel	3374.80	5	Masonry	2	1928
Vivekanand hostel	2953.44	5	Masonry	2	1930
ABLT	620.195	11	Masonry	2	2014
IIT(BHU) main library	2585.21	11	Masonry	2	1932
Training and placement cell	652.362	11	Masonry	2	1932
Lecture theatre complex	825.781	11	Masonry	4	2016

In building type: 5 signifies residential building and 11 for administrative

$$M\ddot{U} + KU = -M\ddot{g} \tag{3}$$

This is governing equation of the mass-spring model, where, $M = \int_{\Omega} \rho\phi^1.\phi^1 dv$, $K = \int_{\Omega} \nabla\phi^1 : c : \phi^1 dv$. This U is solved for a given g with initial condition of $U = 0$ and $\dot{U} = 0$. This mass spring model has been further extended to multi-degree of freedom system to analyze multi-story structures in IES.

Automated Construction of Digital Model

Current version of automated construction of digital model utilizes the building outline (edge information) with height of each structure, or 3D maps, and data on construction material data and structure system (architectural plan). There is a challenge in integrating these data because these data are sourced from numerous autonomous agencies and are stored at different spatial resolution. In nutshell, data are scatters and therefore cannot be used in constructing the analysis model. Moreover, 3D maps of Indian cities are not accurate.

Authors adopted open street maps to generate shape files for extracting building outlines. Once the outlines are generated, attribute table is populated with required addition information including age of structure, nature of buildings (office, residential or industry), building structure (RCC, masonry, steel, etc), building floor (height of building), etc. To generate the digital model of selected site, a shapefile has been created which contains the outlines of all structures (see Fig. 1) and attribute table (see Table 1) using preexisting open street maps on QGIS software.

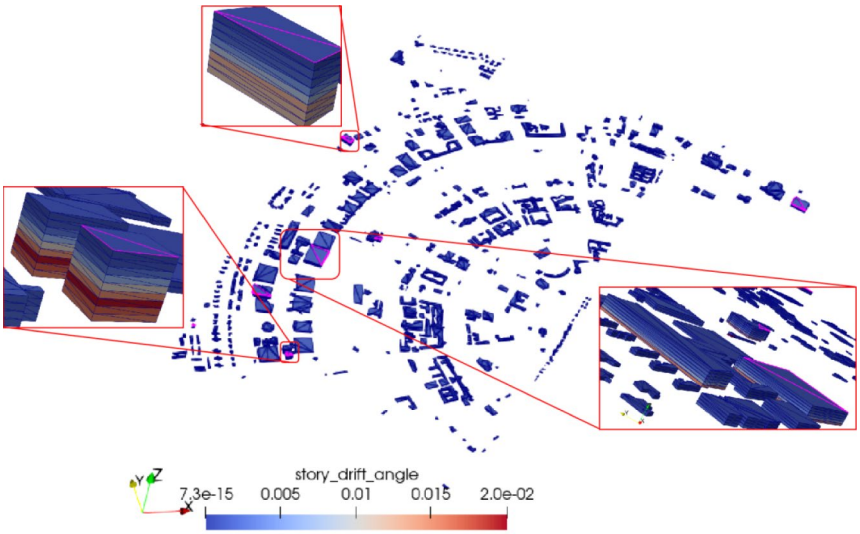
Results and Discussion

In this study, three catastrophic earthquakes- (a) 1995 Kobe Earthquake, (b) 2001 Bhuj Earthquake and (c) 2015 Gorkha (Nepal) Earthquake have been considered for risk assessments due to their widespread destruction of buildings and lifeline infrastructures.

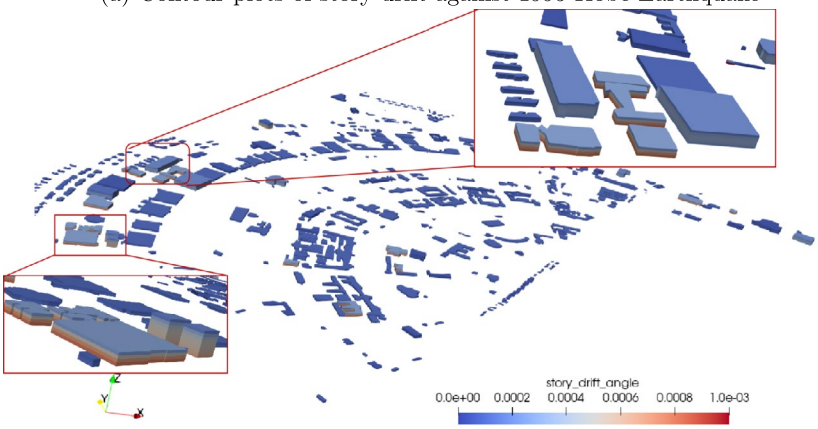
Seismic Risk of BHU Campus

A seismic response analysis of the BHU digital twin was conducted based on the 1995 Kobe earthquake and the 2001 Bhuj earthquake. In the case of the 1995 Kobe earthquake, the faculty building (G+6) on campus exhibited the maximum displacement of 0.55 m and a story drift angle of 5.5%. The contour plots of story drift (see Fig. 2a) indicate that story drift values are higher for all buildings with a height of G+4 or greater. Similarly, during the 2001 Bhuj

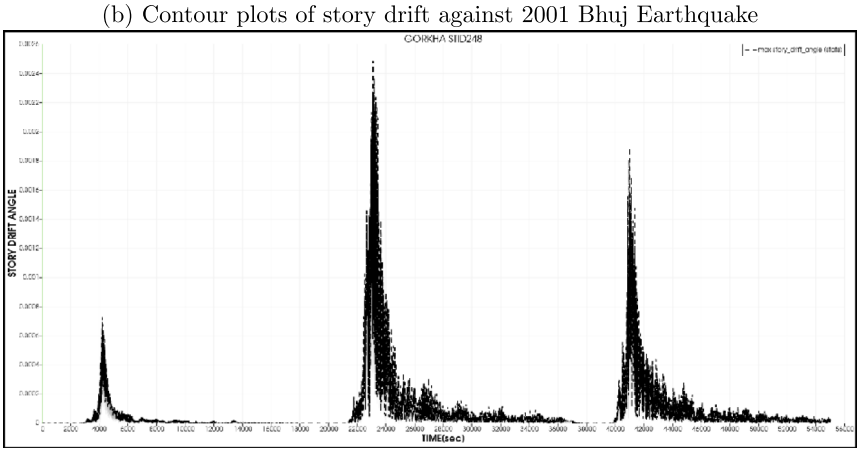
Fig. 2 Seismic response analysis results of BHU campus



(a) Contour plots of story drift against 1995 Kobe Earthquake



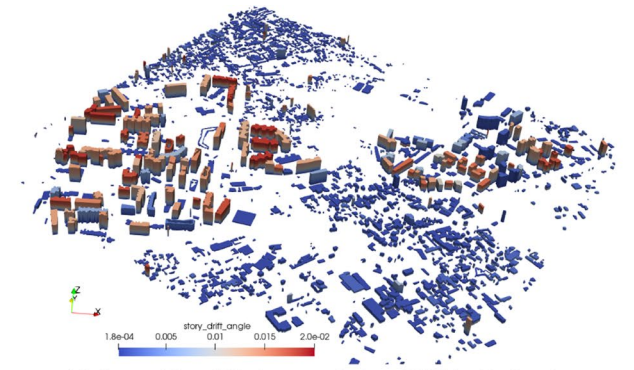
(b) Contour plots of story drift against 2001 Bhuj Earthquake



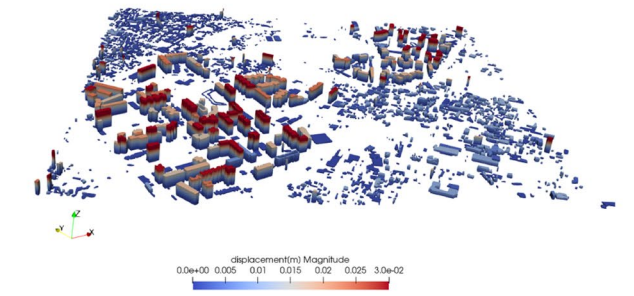
(c) Time history variation of story-draft of a G+6 buildings during 2015 Nepal Earthquake

earthquake, the faculty building again showed the highest values for rooftop displacement and story drift angle, measuring 0.36 m and 0.0035%, respectively (refer to Fig. 2b). Although these values remain within permissible limits, the time-history results demonstrate prolonged vibrations in the buildings.

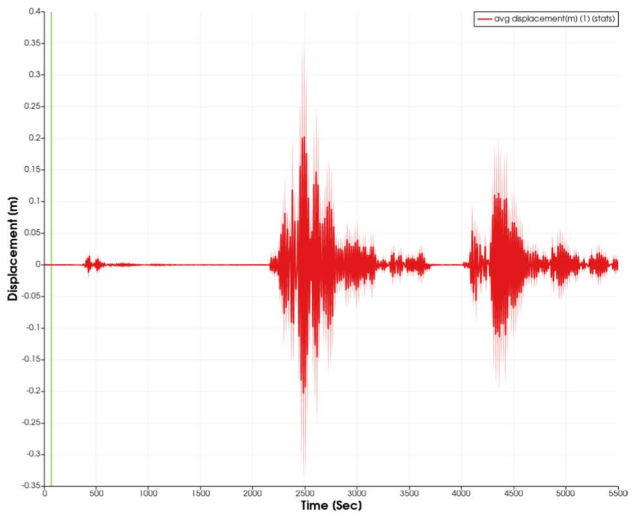
In the simulation results for the 2015 Nepal earthquake, buildings with a height of 15 m or more exhibited amplified story drift (see Fig. 2c) and displacement during the second wave, followed by a reduction in the third consecutive wave. Structures experienced partial damage during the first wave, leading to a reduction in their stiffness and



(a) Contour Plot of Displacement during 1995 Kobe Earthquake



(b) Contour Plot of Displacement during 2001 Bhuj Earthquake



(c) Time history variation of rooftop displacement of a buildings during 2015 Nepal Earthquake

Fig. 3 Seismic response analysis results of crossing Republik, Ghaziabad

an amplification of earthquake-induced deformation. Conversely, some structures sustained severe damage due to the cumulative effects of the first two waves, resulting in reduced deformation during the third wave.

Seismic Risk of Crossing Republik, Ghaziabad

A seismic response analysis was conducted for Crossing Republik Township using all three selected ground motions. This township is primarily composed of high-rise buildings, typically exceeding 40 m in height. As a result, excessive rooftop displacement poses a significant concern in the event of an earthquake. The seismic response of structures during the 1995 Kobe Earthquake (see Fig. 3a) and the 2001 Bhuj Earthquake (see Fig. 3b) indicates that the rooftop displacement in these high-rise buildings surpasses the permissible limit of $h/500$, where h represents the building height.

The time history plots of horizontal displacement are plotted in Fig. 3c during the 2015 Nepal Earthquake. From the plots, it is clear that the amplitude of displacement amplified from the first wave to the second and then reduces in the third wave, suggesting that the structure is probably already damaged in the second wave. Furthermore, it is also observed that prolonged vibrations of higher amplitudes are present in the second wave. The second wave is very identical to the 2001 Bhuj Earthquake and shows similar results. However, the third wave has become catastrophic, and most of the structures have experienced large deformation (see Fig. 3c), suggesting severe damage. These values suggest that buildings higher than 40 m exhibit maximum story drift in the third wave of ground motion, and all other structures experience maximum drift in the second wave, which persists in the third wave.

In a hypothetical scenario where the BHU campus or Crossing Republik township experienced an earthquake comparable to the 1995 Kobe earthquake, structures would likely have suffered significant damage. Although structures on the BHU campus might have demonstrated better resilience during the 2001 Bhuj earthquake, they would still have incurred considerable non-structural damages. It's important to consider that the ground motion data from the 2001 Bhuj earthquake, used in this study, was recorded 400 km away from the epicenter. As a result, the recorded responses may be on the lower side. Additionally, had the BHU campus or Crossing Republik township faced an earthquake similar to the magnitude of the 2015 Gorkha earthquake, many buildings taller than 15 m would have sustained damage or experienced partial collapse.

Summary and Future Work

A digital twin of BHU campus and Crossing Republik, Ghaziabad has been constructed and analyzed for 1995 Kobe Earthquake, 2001 Bhuj Earthquake and 2015 Nepal Earthquake. The digital models have been constructed by utilizing Open Street Maps and attribute table feature of QGIS, which enlist various attributes such as building height, type of building, and nature of buildings. preliminary results suggest that story drift of mid-rise structures, height exceeding 15 m, is beyond limiting safe values of $0.01h$; h being building height, inferring the higher seismic risk. This research

has been limited to a cluster of structures inform of a university campus or group housing society. However, these study prevails in demonstrating the capabilities of IES in seismic risk assessment beyond Japanese cities and highlighting the underline challenges in its adaptation for Indian cities.

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